

Second and third party punishment under costly monitoring



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ABSTRACT

In a laboratory experiment we study how costly punishment behavior of second and third parties in a social dilemma situation is affected by monitoring costs. Subjects have to pay a fee over and above punishment costs if they wish to condition punishments on previous play, which is equivalent to a binary choice between the acquisition of perfect information on the target subject's behavior and no information at all. When monitoring is costly both second and third party punishment is weaker and less discriminate and hence generates weaker incentives for cooperation than when monitoring is free. There are subtle differences between second and third parties: The presence of monitoring costs leads subjects to withhold sanctioning more often as second parties than as third parties, and to punish indiscriminately more often as third parties than as second parties. The results contribute to the understanding of peer-enforcement of cooperation in social dilemmas and whether there is a common motivational structure underpinning second and third party punishment.

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1. Introduction

The resolution of social dilemmas can be greatly facilitated by (costly) punishment of non-cooperative behavior by vested «second parties» or by-standing «third parties» (e.g. Bowles & Gintis, 2011; Gintis, Bowles, Boyd, & Fehr, 2005; Henrich et al., 2004). However, in many important settings such persons do not immediately and perfectly observe whether the target subject cooperated or defected. In such situations they can either decide to mete out punishments under uncertainty, bearing the risk of error,¹ or remedy the lack of information before deciding to punish, typically at a cost. We call such remediation «monitoring». Monitoring effort to overcome imperfect information on coplayers' actions is well known in a variety of economically relevant contexts: For example, community-based management of common-pool resources relies on time- and effort-consuming peer-monitoring activities (Ostrom, 1990), such as conducting patrols through a local forest in order to detect violations of the applicable stewardship norms (Rustagi, Engel, & Kosfeld, 2010). Costly monitoring activities are also documented in production teams and cooperatives (Acheson, 1975; Craig & Pencavel, 1995; Dong & Dow, 1993;

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¹ There are two types of error a second or third party might perform under uncertainty (relative to her own preferences): First, she could punish a person that she would not like to punish under certainty. Second, she could let a person escape unpunished that she would like to punish under certainty.

Kandel & Lazear, 1992; Palmer, 1991), labor relations (Kanemoto & MacLeod, 1991; Lazear, 1993; Shapiro & Stiglitz, 1984), finance (Armendáriz & Morduch, 2005; Williamson, 1987), or neighborhood watch (Sampson, Raudenbush, & Earls, 1997).

In the present paper we implement such a setting in a laboratory experiment and investigate how punishment behavior of second and third parties is affected by monitoring costs. We draw on a well-known paradigm by Fehr and Fischbacher (2004b, FF in what follows) that combines a one-shot prisoners' dilemma (PD) with a costly punishment stage. Each subject plays the game in two conditions, a second party punishment condition (2PP) and a third party punishment condition (3PP), with randomly matched and unknown coplayers. In the first stage, all subjects decide between «cooperate» or «defect». Subsequently, in the 2PP both players in each group can choose to punish the other. In the 3PP, each player has the opportunity to punish a player from another group. Punishment is costly to both the punishing and the punished subject, yet three times as costly for the latter compared to the former. Punishment decisions are elicited through the *strategy method* (SM): subjects specify a complete punishment strategy, i.e. a punishment level for each possible first-stage action profile, without being informed about the actual profile. Consistent with the SM, we implement monitoring *implicitly*: In FF, the space of punishment strategies is unrestricted in a way that players eligible to punish can act *as if* they were perfectly informed about the target subject's first-stage action. We require subjects to pay a fee over and above punishment costs if they wish to condition punishments on previous play, which is equivalent to a binary choice between the acquisition of perfect information on the target subject's behavior («perfect monitoring»), and no information at all («no monitoring»). By this design we elicit (i) individual-level data on subjects' associated punishment *strategies* (instead of single actions), and (ii) isolate the *instrumental* motivation to monitor, i.e. for the purpose of conditioning punishment on the history of play, while ruling out experimenter demand effects, general curiosity, distrust vis-à-vis the experimenters or related motivations.²

We implement three treatment conditions with different levels of monitoring costs, including zero as a baseline condition. The latter constitutes a replication of FF. We find that both second and third party punishment is less discriminate and hence generates weaker incentives for cooperation when monitoring is costly compared to a setting in which monitoring is free. Specifically, the fraction of monitoring subjects is substantially smaller in the former than in the latter, and among the non-monitoring subjects the vast majority does not punish at all, whereas there is also a significant fraction of «blind punishers» that punish unconditionally. There are subtle differences between second and third parties: The presence of monitoring costs leads subjects to withhold sanctioning more often as second parties than as third parties, and to punish indiscriminately more often as third parties than as second parties.

Our paper contributes to the experimental literature on peer-enforcement of cooperation in social dilemmas, specifically under imperfect monitoring. We replicate existing evidence that both second and third parties engage in costly non-strategic punishment and most of them target punishments on defectors if doing so is feasible and costless (see e.g. Fehr & Fischbacher, 2004b; Henrich et al., 2004; Leibbrandt & López-Pérez, 2012). Our study differs from this literature in that not only punishment is costly, but also the act of monitoring that allows punishment to be targeted in the first place.³ Apart from adding situational richness to this literature, our study thereby contributes new evidence about commonalities and differences between second and third party punishment behavior. Our study is also related to a literature on exogenously noisy or constrained monitoring in social dilemmas with punishment (Ambrus & Greiner, 2012; Bornstein & Weisel, 2010; Carpenter, 2007). Our study differs from this literature in that monitoring imperfections are not exogenously given, but can be endogenously eliminated.

In the remainder of the paper we proceed as follows. In Section 2 we describe the experiment and state the expected treatment effects. We then present the results in Section 3 and conclude in Section 4.

2. Experiment

We begin by describing the basic game design, which is essentially identical to FF (Section 2.1). We then introduce some formalization in order to demonstrate that imperfect observation of first-stage actions can be mimicked by constraints on the space of punishment strategies, present the implementation of the monitoring and punishment stage, and briefly comment on the advantages of this design (Section 2.2). Finally, we describe the treatment conditions and state the expected treatment effects (Section 2.3), and report the experimental procedures (Section 2.4).

2.1. Basic game design

Each subject played a game consisting of a one-shot PD stage and a costly punishment stage in two conditions (both were paid), a second party punishment condition (2PP) and a third party punishment condition (3PP), with randomly matched and

² While monitoring costs are the most natural interpretation of the design, the implementation is conceptually consistent with other constraints that make conditioning punishment on previous play costly. We are grateful to an anonymous referee for raising this point. Such constraints could be structural limitations of the punishment technology or institutional constraints such as procedural rules in court that require specific evidence for specific types of punishment to be imposed, necessitating further effort to be spent on additional investigations.

³ Two experiments using the repeated public goods game paradigm also introduce costly information acquisition about individual contributions: Samek and Sheremeta (2016) finds that such information is rarely demanded, and Kurzban and DeScioli (2008) finds that observation costs have a contribution-decreasing effect. We thank an anonymous referee for pointing us to these papers. In both papers, there is no punishment feasible and that there are strategic reasons for information acquisition. This contrasts with our study of one-shot interactions with punishment, in which the demand for information is not productive for informing strategic choice.

unknown coplayers in random sequence. In the first stage, all subjects could decide to «cooperate» by transferring their endowment of 10 tokens, in which case the experimenter tripled it, or to «defect» by keeping their endowment for themselves. The payoffs are tabulated in Fig. 1a.

At the beginning of the second stage, each player received an additional endowment of 40 tokens in both conditions. Subsequently, in the 2PP both players in each group could choose to punish the other (see the solid arrows in Fig. 1b). In the 3PP, the first (second) player in a group had the opportunity to punish a player from another (a third) group (see the dashed arrows in Fig. 1b).⁴ Thus, in both the 2PP and the 3PP, respectively, each player was eligible to punish exactly one other player. A linear sanctioning technology was used⁵: any token spent by the punishing player subtracted three tokens from the punished player's payoff. Expenditures were restricted to integers and a maximum of 20 tokens.

Subjects reported their punishment decisions in the same way as in FF: in the 2PP players announced a contingent punishment strategy, indicating a punishment level for each of the target player's possible transfer decisions, without being informed about the actual decision. Likewise, in the 3PP each third party, while being informed about the first-stage decision of the other member in their own group, had to indicate a punishment strategy over four possible contingencies. We now represent this a bit more formally and then introduce the novel element of (costly) monitoring of the target player's first-stage action.

2.2. Monitoring

We take the outcome of the first stage as given and model the second stage as a decision problem based on the state-space model of information (Aumann, 1976; Hiniikka, 1962). There are three players called A (the «first party»), B (the «second party») and C (the «third party»). Let $\omega \equiv (\omega_A, \omega_B)$ be the action profile from the PD played by A and B in the first-stage, where $\omega_i \in \{c, d\}$, c stands for «cooperate» and d for «defect». We consider the second stage from the perspective of person j that is eligible to punish person A, that is, $j = B$ in the 2PP and $j = C$ in the 3PP. The four possible first-stage action profiles constitute the state space $\Omega \equiv \{(d, d), (d, c), (c, d), (c, c)\}$, i.e. the possible worlds in which j can find herself.

An *information correspondence* for j is a map $F_j : \Omega \rightarrow 2^\Omega \setminus \emptyset$ that associates to each state $\omega \in \Omega$ a non-empty subset $F_j(\omega) \subseteq \Omega$, called j 's *information set* in state ω , that represents j 's informedness in that state: each state $\omega \in F_j(\omega)$ is considered possibly true by j , and each state $\omega \in \Omega \setminus F_j(\omega)$ is certainly false. We impose the standard assumption that F_j is *partitional*, that is, each information function induces a partition $\mathcal{F}_j$ of the state space Ω .⁶

Let P_j be the set of punishment levels j is eligible to impose on A. Player j 's task is to choose a function $p_j : \Omega \rightarrow P_j$, called a *punishment strategy*, for which $p_j(\omega) = p_j(\omega')$ whenever $\omega \in F_j(\omega) \wedge \omega' \in F_j(\omega)$. That is, j needs to specify a punishment level for each possible state under the constraint that she cannot choose different punishment levels in states that are indistinguishable to her. Thus, j 's degree of uninformedness imposes *restrictions* on the set of feasible punishment strategies.

FF allow the specification of a different punishment level for each possible first-stage action performed by A: In the 3PP, $p_C(\omega)$ is specified for each $\omega \in \Omega$ and all four punishment levels are allowed to be different. In the 2PP $p_B(\omega)$ is specified with the restrictions $p_B(d, d) = p_B(d, c)$ and $p_B(c, d) = p_B(c, c)$, that is, B chooses a punishment level for each of A's two possible first-stage actions.⁷

We now demonstrate that FF's implementation of the SM is equivalent to a naturally (i.e. by means of the direct-response method, DRM) played version of the game in which A's first-stage action can be *perfectly monitored* by j (i.e. j receives a perfect signal on A's first-stage action before imposing punishments). To see this, define $S : \Omega \rightarrow 2^\Omega \setminus \emptyset$ that associates to each state $\omega \in \Omega$ a subset $S(\omega) \subseteq \Omega$ with $\omega \in S(\omega)$, called *signal*. Thus, any state produces a signal $S(\omega)$ to j that contains the true state ω and possibly other states. All states in $S(\omega)$ are possible while all states in $\Omega \setminus S(\omega)$ can be ruled out.⁸ A signal is called *perfect* if it is a singleton that contains only the true state, $S(\omega) = \{\omega\}$. Otherwise it is called *imperfect* with the boundary case $S(\omega) = \Omega$ that can be interpreted as «no signal». Because F_j is partitional and $S(\omega)$ contains the true state, j deduces that the true state must be in both $F_j(\omega)$ and $S(\omega)$. Thus, for all $\omega \in S$ we have $\hat{F}_j(\omega) = F_j(\omega) \cap S(\omega)$, where $\hat{F}_j$ is j 's updated information correspondence.⁹

Now, assume that j receives a perfect signal at the beginning of the second stage, before she metes out punishments. Then $\hat{F}_j(\omega) = \{\omega\}$ for all $\omega \in \Omega$, or equivalently

⁴ This matching protocol rules out «tacit collusion» between third parties (see Fehr & Fischbacher, 2004b).

⁵ See Casari (2005) on potential biases due to non-linear sanctioning technologies.

⁶ This is equivalent to the assumptions that (i) $\omega \in F_j(\omega)$ holds for every $\omega \in \Omega$ (j cannot be convinced of things that are false) and (ii) $\omega' \in F_j(\omega) \Rightarrow F_j(\omega') = F_j(\omega)$ (j uses the consistency or inconsistency of states with her information to make inferences about the state). See for example Osborne and Rubinstein (1994).

⁷ Here FF's implementation of the SM is not entirely rigorous: theoretically, a fully specified strategy would allow B also to condition her punishments on her own first-stage action. However, because we want to replicate FF's study as a baseline as close as possible, we follow their application.

⁸ Put differently, any subset $E \subseteq \Omega$ is called an *event*. Event E occurs in state ω if and only if $\omega \in E$. Conversely, if E occurs the true state must be in E . The signal $S(\omega)$ is an event that occurs in ω .

⁹ A signal in state ω is *uninformative* if $F_j(\omega) \subseteq S(\omega)$ because then $\hat{F}_j(\omega) = F_j(\omega)$, otherwise it is *informative*. In other words, a signal is informative if it allows j to rule out at least one state that she previously considered possible. Note that $F_j(\omega)$ and $S(\omega)$ can never be disjoint because they both contain the true state. Thus, a perfect signal is only uninformative if j already knew the true state. Because $\hat{F}_j(\omega) \subseteq F_j(\omega)$ holds for all $\omega \in \Omega$, the induced information partition $\mathcal{F}_j$ is never *coarser* (less and bigger information sets) than $\mathcal{F}_j$, representing the fact that j cannot be less informed after receiving $S(\omega)$.

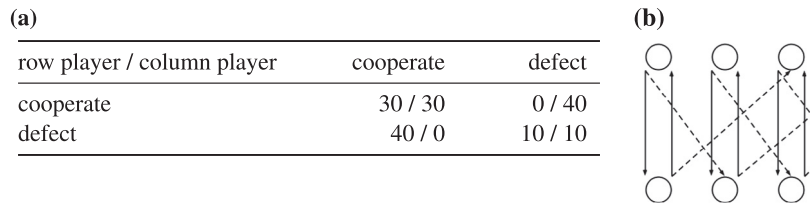


Fig. 1. The PD in the first stage (panel a) and the second-stage matching (panel b) in the 2PP (solid arrows) and the 3PP (dashed arrows).

$$\hat{\mathcal{F}}_j = \{\{(d, d)\}, \{(d, c)\}, \{(c, d)\}, \{(c, c)\}\}$$

But this implies that there is no pair of states that is in the same information set, such that the set of feasible punishment strategies is not restricted by the information structure. Specifically, in the 3PP player C is free to choose a different punishment level $p_C(\omega)$ in each of the four states. In the 2PP player B is free to specify a different punishment level for each of A's two possible first-stage actions. This demonstrates that FF's implementation is *equivalent* to a naturally played (or DRM-played) game with perfect monitoring of A's (and B's) first-stage actions.

We can also show that this equivalence of information structures and constraints on the strategy space naturally extends to cases with imperfect signals, including the case of no signal. Specifically, the imposition of the restriction $p_j(\omega) = p_j(\omega')$ for some $\omega \in \Omega$ and $\omega' \in \Omega$ effectively mimics a particular information structure in which ω and ω' belong to the same information set. In the present paper, we are interested in the two extremes, perfect ignorance and perfect information about player A's first-stage action. The latter is characterized formally above. Perfect ignorance is implemented in the 2PP by the restrictions $p_B(c, a_B) = p_B(d, a_B)$ for any $a_B \in \{c, d\}$. Those constraints on the set of punishment strategies are identical to the ones imposed by an information partition

$$\hat{\mathcal{F}}_B = \{\{(d, d), (c, d)\}, \{(d, c), (c, c)\}\}$$

i.e. B knows her own but not A's first-stage action. Together with the restrictions $p_B(a_A, d) = p_B(a_A, c)$ for any $a_A \in \{c, d\}$ (see above) this implies that $p_B(\omega)$ must be identical for all $\omega \in \Omega$. Likewise, in the 3PP the restrictions $p_C(c, a_B) = p_C(d, a_B)$ for any $a_B \in \{c, d\}$ on the set of punishment strategies are identical to the ones imposed by an information partition

$$\hat{\mathcal{F}}_C = \{\{(d, d), (c, d)\}, \{(d, c), (c, c)\}\}$$

i.e. C knows B's but not A's first-stage action.

Wrapping up, a decision to choose between a punishment strategy with or without the specified restrictions is equivalent to a decision between receiving a perfect signal and receiving no signal at all about A's first-stage action. In other words, we can mimic a monitoring decision within the SM (and thus without actually informing j about A's first-stage action) by allowing subjects to choose between a punishment strategy in which punishments are conditioned on a_A and a strategy in which punishments are *not* conditioned on a_A . We term the former strategy *discriminate* and the latter *indiscriminate*. Choosing a discriminate punishment strategy is equivalent to monitoring the target player before the application of punishments.¹⁰

For later reference, we define the following three «punishment types»: First, a player that decides not to punish at all, i.e. $p_j(\omega) = 0$ for all $\omega \in \Omega$, is referred to as type *N* (for «non-punisher»). Among the punishing subjects ($p_j(\omega) > 0$ for some $\omega \in \Omega$), there are two types. «Blind punishers» punish defectors and cooperators to the same extent, that is, they choose an indiscriminate strategy. This type is denoted as *B*. The other type *T* are «targeted punishers» that choose a discriminate strategy. The *B*-types and *T*-types together form the class of «punishers», denoted *P*.

Why is it useful to use this indirect implementation of monitoring using the SM instead of using a simple DRM-design? First, the SM is more efficient. If we denote the true state $\bar{\omega}$, the DRM elicits only $p_j(\bar{\omega})$, whereas the SM elicits the complete strategy $p_j(\omega)$. In the present study this is particularly useful because it permits the analysis of individual-level punishment *strategies*, which allow for deeper insights than single actions.¹¹ For instance, the above classification of types would not be possible with DRM-data.

Second, this method cleanly isolates the *instrumental* motivation to monitor, i.e. the acquisition of information for the purpose of conditioning punishments on first-stage actions. In a DRM-design in which subjects would have to explicitly demand information (e.g. by clicking a button), monitoring could be driven by a variety of other factors, some of which one would clearly classify as a confound. For example, the mere presence of an explicit decision about getting information (e.g. the presence of a button) could induce experimenter demand effects («if there is a button, then I should click it»). It could also measure general curiosity or distrust vis-à-vis the experimenters. Within our more indirect SM-design there is no button to click, and no actual information provided—in fact the decision-situation is in essence identical to the one in

¹⁰ On this basis it is also easy to see that monitoring costs are *not* the same as punishment costs: monitoring costs are incurred *independent* from the realization of the state while the level of punishment costs depends on the state.

¹¹ Under the DRM, observing a subject to spend x tokens on punishing a defector is not informative about that subject's preference to condition punishments on previous play, because the counter-factual case in which the target subject cooperates is not observed.

FF's experiment. This rules out the above kinds of motivations and cleanly isolates the instrumental motivation behind monitoring.¹²

2.3. Treatment conditions and predictions

In our experiment, a fee over and above punishment costs fell due for any discriminate punishment strategy. We varied the level of this fee between-subjects at three levels: A fee of zero is an obvious starting point—it provides a direct replication of FF and therefore a meaningful baseline treatment. In what follows, this treatment is labeled *M0*. Given the payoff structure of the game, we expected a fee of ten tokens being close to prohibitive such that we have chosen this level as a reasonable boundary of the considered fee interval (treatment *M10*). We implemented another condition at the midpoint of the fee interval at five tokens (treatment *M5*).

In the baseline condition (*M0*) we expect to replicate the key results of FF. Furthermore, we expect all punishing subjects to behave as *T*-types, i.e. to specify a different punishment level for a defector than for a cooperator. This is because in *M0* discriminate and indiscriminate punishment strategies have the same price (zero) such that an arbitrarily weak preference to target punishment suffices to opt for the former.

Since the price of discriminate punishment strategies relative to the other punishment strategies is higher in the costly monitoring conditions than in the baseline, we expect that the frequency of *T*-types will be lower in the costly monitoring conditions compared to the baseline condition in both the 2PP and the 3PP.¹³ Conversely, we expect that the frequency of *N*-types and *B*-types to be higher under costly monitoring, but the latter to remain the exception.¹⁴ As a direct consequence, the incentive to cooperate, defined as the difference between average punishment of defection and average punishment of cooperation, should be lower in the costly monitoring conditions compared to the baseline.

Finally, there are various plausible considerations with respect to whether and how second and third parties might differ in their responses to monitoring costs. On the one hand, since third parties are not materially affected they might be less interested in the target subject's first-stage behavior than second parties, such that a given raise of the monitoring fee could «crowd out» relatively more third parties than second parties from the class of *T*-types, and consequently have a stronger (negative) effect on the incentive to cooperate in the 3PP than in the 2PP.¹⁵

Alternatively, third parties are often suspected to be less egocentric, more impartial, and motivated more strongly by morality and ethical norms than second parties (Fehr & Fischbacher, 2004b; Fehr & Fischbacher, 2004a; Fehr, Fischbacher, & Gächter, 2002). This implies that a given raise of the monitoring fee should «crowd out» relatively more second parties than third parties from the class of *T*-types, and consequently have a stronger (negative) effect on the incentive to cooperate in the 2PP than in the 3PP.

Finally, there is behavioral (Leibbrandt & López-Pérez, 2012) and neuroscientific evidence (Buckholtz et al., 2008) suggesting that the motivational structures underlying second and third party punishment is not significantly different.¹⁶ If this is true, then we should observe no significant differences between second and third parties' responses to monitoring costs.

2.4. Procedures

Participants were recruited from the general undergraduate student population using the online recruitment system ORSEE (Greiner, 2004). In total 134 subjects participated,¹⁷ and under consideration of experimental efficiency (e.g. List, Sadoff, & Wagner, 2011; McClelland, 1997) we allocated approximately half of them to the midpoint cell *M5* (66 subjects), a third to maximum cell *M10* (46 subjects), and a sixth to the baseline cell *M0* (22 subjects), respectively. Of course, no subject participated in more than one session and monitoring fee treatment.

The experiment was conducted with a personal computer network and the software z-Tree (Fischbacher, 2007) at the experimental laboratory of the Alfred-Weber-Institute (AWI-Lab) at the University of Heidelberg. Upon entering the

¹² But note that behavior in structurally identical games may differ depending on whether the SM or the DRM is used. For example, the emotional load of a decision may be greater under the DRM («hot decision») than under the SM («cold decision»). However, the differences are well predictable (Brandts & Charness, 2011).

¹³ Punishment behavior has been shown to be elastic with respect to incentives: increasing the cost of punishment generally reduces the supply of sanctions (Anderson & Putterman, 2006; Carpenter, 2007; Egas & Riedl, 2008; Oosterbeek, Sloof, & van de Kuilen, 2004; Suleiman, 1996).

¹⁴ Consider a player with a preference to target punishments on defectors. Under costless monitoring this player will behave as type *T*. Now, if a given monitoring fee renders the price of a discriminate punishment strategy prohibitive to this player, then refraining from punishment altogether (type *N*) appears to be the intuitively natural response. However, there is also evidence on pure «money burning» or « nastiness» (Abbink & Herrmann, 2011; Abbink & Sadrieh, 2009; Zizzo, 2003; Zizzo & Oswald, 2001), that is, costly «punishment» without any history of previous interaction on which it could be conditioned upon. The latter implies *B*-type behavior under costly monitoring.

¹⁵ FF found that cooperators were punished, on average, equally strong by second and third parties, respectively, but defectors were punished markedly stronger by second parties compared to third parties. This suggests that the average second party might have a higher reservation price for discriminate strategies (or a stronger aversion to place punishments indiscriminately) than the average third party. A motivational underpinning could be that direct reciprocity (e.g. Dufwenberg & Kirchsteiger, 2004; Falk & Fischbacher, 2006; Fehr & Gächter, 1998; Goranson & Berkowitz, 1966; Gouldner, 1960; Rabin, 1993) can be a concern for second parties but not for third parties, because there is no history of previous interaction between a third party and the target subject.

¹⁶ The evidence on this issue is not conclusive, however (see e.g. Carpenter & Matthews, 2009; Fehr & Fischbacher, 2004b).

¹⁷ Among the participants 49.3 percent were female and a share of 64.2 percent had never participated in a laboratory experiment before. The mean age was 21.8 years.

Table 1
Intensity and frequency of second and third party punishment by treatment condition.

Target player is	TP punishment Target player's coplayer is								
	SP punishment			Cooperator			Defector		
	M0	M5	M10	M0	M5	M10	M0	M5	M10
Defector	7.6 (8.4) <i>72.7%</i> (66.7%)	3.3	1.9	6.0 (3.1) <i>72.7%</i> (58.7%)	2.6	2.8	2.3 (1.4) <i>45.5%</i> (32.6%)	1.6	1.7
Cooperator	0.6 (0.7) <i>13.6%</i> (8.3%)	1.1	0.3	1.4 (0.6) <i>22.7%</i> (15.2%)	1.3	1.5	1.1 (0.7) <i>22.7%</i> (15.2%)	1.0	0.6
Defector = Cooperator	.0001	.0063	.0144	.0002	.0009	.0054	.0988	.0143	.0082
Defector $\perp$ fee		←.0002→			←.0051→			←.2909→	
Cooperator $\perp$ fee		←.7563→			←.5995→			←.7955→	

Average (per subject) expenditure for punishment points in regular letters. Relative frequency of punishing subjects in italics. The results of Fehr and Fischbacher (2004) are in parentheses. The third row from below reports the p -values of Wilcoxon signed-rank tests with the H_0 that cooperators and defectors are punished equally strong in the respective column. The last two rows report the p -values of Kruskal-Wallis tests with the H_0 that the punishment of defectors (second row from below) and cooperators (last row) is independent from the monitoring fee.

laboratory, subjects were randomly assigned to the computer terminals. Direct communication among them was not allowed for the duration of the entire session. Booths separated the participants visually, ensuring that they made their decisions anonymously and independently. Furthermore, subjects did not receive any information on the personal identity of any other participant, neither before nor while nor after the experiment.

At the beginning of the experiment, that is, before any decisions were made, subjects received detailed written instructions that explained the exact structure of the game and the procedural rules. The experiment was framed in a sterile way using neutral language and avoiding value laden terms in the instructions (see [supplementary material](#)), which were identical in all conditions except the fee level. Participants had to answer a set of control questions individually at their respective seats in order to ensure comprehension of the rules. We did not start the experiment before all subjects had answered all questions correctly.

Since each subjects played both the 2PP and the 3PP we counter-balanced the order of the conditions (Foley, 2004), i.e. in half of the sessions the 2PP was played before the 3PP, in other half the 3PP before the 2PP.¹⁸ The assignment of subjects to those sequences was random. In addition, subjects were not informed about the second game before the end of the first game. The exact timing of events was as follows. First, the subjects were randomly matched into groups of two. Then each subject made her or his decisions in the first game. After being informed about the payoffs in the first game, the experimenter announced that a second experiment will be conducted and distributed additional instructions that explained the differences to the first game. After being randomly re-matched into new groups, each subject made her or his decisions in the second game. After being informed about the payoffs, subjects were asked to answer a short questionnaire while the experimenter prepared the payoffs. Subjects were then individually called to the experimenter booth, payed out (according to a random number matched to their decisions; no personal identities were used throughout the whole experiment) and dismissed. The whole experiment lasted approximately 90 minutes and subjects earned an average €18.46 (€0.10 per token earned), including a fixed show-up fee. Earnings exceed the local average hourly wage of a typical student job and can hence be considered meaningful to the participants.

3. Results

We begin by presenting the results from the baseline that replicates the study by FF. Against this benchmark, we then describe the core results of the experiment.

The aggregate results from the baseline condition are shown in the M0-columns of Table 1. For the readers convenience we repeat FF's results in parentheses. In line with FF, a clear majority of both second and third parties punish defectors whereas punishment of cooperators is present but infrequent and weak. The prediction that second party punishment is stronger than third party punishment is, in line with Leibbrandt and López-Pérez (2012), replicated only partially. Table 1 shows that second parties spent more to punish defectors on average, but the difference is statistically significant only in case the target subject's coplayer in the 3PP also defected. This is due to the fact that third party punishment, especially

¹⁸ If there would be effects of order they can thus be expected to be eliminated (or at least mitigated) in the pooled dataset (Foley, 2004, see also Gächter & Herrmann, 2010; Resnick, Zeckhauser, Swanson, & Lockwood, 2006; Tan & Vogel, 2008; Zizzo & Fleming, 2011, for applications in experimental economics). However, there is no evidence in favor of such effects anyway and all key results of this paper also hold under each sequence individually (see Appendix B).

Table 2

«Transition matrix» of behavioral types across the 2PP and the 3PP.

2PP type		3PP type			
		<i>N</i>	<i>T</i>	<i>B</i>	Margin
<i>N</i>	<i>M0</i>	18.2(4)	9.1(2)	0.0(0)	27.3(6)
	<i>M5</i>	56.1(37)	4.5(3)	6.1(4)	66.7(44)
	<i>M10</i>	54.3(25)	6.5(3)	13.4(6)	73.9(34)
					.0007 ^a
<i>T</i>	<i>M0</i>	9.1(2)	59.1(13)	4.5(1)	72.7(16)
	<i>M5</i>	3.0(2)	10.6(7)	7.6(5)	21.2(14)
	<i>M10</i>	0.0(0)	6.5(3)	6.5(3)	13.4(6)
					.0001 ^b
<i>B</i>	<i>M0</i>	0.0(0)	0.0(0)	0.0(0)	0.0(0)
	<i>M5</i>	3.0(2)	1.5(1)	7.6(5)	12.1(8)
	<i>M10</i>	2.2(1)	4.3(2)	6.5(3)	13.4(6)
					.2152 ^c
Margin	<i>M0</i>	27.3(6)	68.2(15)	4.5(1)	100(22)
	<i>M5</i>	62.1(41)	16.7(11)	21.2(14)	100(66)
	<i>M10</i>	56.5(26)	17.4(8)	26.1(12)	100(46)
		.0171 ^a	.0001 ^b	.1136 ^c	

Shown are the relative frequencies (in percent) and the absolute frequencies in parantheses. Reported in italics are the *p*-values of Kruskal–Wallis tests with the H_0 that the marginal frequencies of the respective types are independent from the monitoring fee. (a) Pairwise, the differences between *M0* and *M5* (Mann–Whitney tests, $p = .0013$ in the 2PP and $p = .0048$ in the 3PP), and *M0* and *M10* ($p = .0003$ in the 2PP and $p = .0248$ in the 3PP) are significant, the differences between *M5* and *M10* are not ($p = .4140$ in the 2PP, $p = .5539$ in the 3PP). (b) Pairwise, the differences between *M0* and *M5* ($p = .0000$ in the 2PP and the 3PP), and *M0* and *M10* ($p = .0000$ in the 2PP and the 3PP) are significant, the differences between *M5* and *M10* are not ($p = .2689$ in the 2PP, $p = .9203$ in the 3PP). (c) In the baseline the frequency of *B*-types is not significantly different from zero (Wilcoxon test, $p = .3173$ in the 3PP; in the 2PP the frequency is strictly zero), whereas in the costly monitoring conditions they are ($p = .0047$ in the 2PP–*M5*, $p = .0002$ in the 3PP–*M5*, $p = .0143$ in the 2PP–*M10*, and $p = .0005$ in the 3PP–*M10*). Pairwise, the differences between *M0* and *M5* ($p = .0886$ in the 2PP and $p = .0734$ in the 3PP), and *M0* and *M10* ($p = .0782$ in the 2PP and $p = .0359$ in the 3PP) are marginally significant, the differences between *M5* and *M10* are not ($p = .8851$ in the 2PP, $p = .5495$ in the 3PP).

of unilateral defectors, is stronger in our experiment than in FF.¹⁹ Anyway, most important for our purposes here is that all punishing second parties (16 out of 16) and all but one punishing third party (15 out of 16) behaved as type *T* in the baseline condition.

Both second and third parties respond to monitoring costs as predicted. Consider the distributions of our «punishment types» shown at the margins of Table 2, where the column margin represents the 2PP and the row margin the 3PP (ignore the interior cells of the table for the moment). First, as evident from the *T*-rows and *T*-columns, the share of targeted punishers is clearly smaller in the costly monitoring conditions compared to the baseline in both second and third parties. Second, as shown by the *N*-rows and *N*-columns, there is a higher share of *N*-types in the costly monitoring conditions compared to the baseline condition. Third, while «blind punishers» are virtually absent in the baseline, the *B*-rows and *B*-columns show that among both second and third parties there is a small but clearly positive fraction of *B*-types in the costly monitoring conditions.

Those strategy adjustments leave the predicted footprint in average punishment levels, shown in the *M5* and *M10* columns of Table 1. On the one hand, defectors are still punished stronger than cooperators on average by both second and third parties in each of the costly monitoring conditions. On the other hand, sanctions are weaker and less directed to defectors compared to the baseline. Hence, defecting got less costly as monitoring costs rise: the incentive to cooperate, as defined above and shown in columns two through four of Table 3, decreases substantially as monitoring gets costly, both in the 2PP and the 3PP. In the 2PP the incentive decreases by 4.8 tokens (69.0 percent) from the baseline to *M5* and by 5.5 tokens (78.3 percent) from the baseline to *M10*. The effective punishment of unilateral defectors in the 3PP decreases by 3.3 tokens (71.6 percent) from the baseline to *M5* and by 3.3 tokens (71.9 percent) from the baseline to *M10*.²⁰

All of the above results apply to both the 2PP and the 3PP. We now turn to some subtle quantitative differences between them. Return to Table 2 first: comparing the frequency of *T*-types between the 2PP and the 3PP in both costly monitoring

¹⁹ A further result of FF was that both cooperators and defectors punished defectors, but cooperators punished more frequently and imposed stronger sanctions. This result, however, was statistically not as solid as the others due to the small number of defectors. Furthermore, it is not of primary interest for our purposes here. The interested reader finds this result replicated in Appendix A.

²⁰ Like FF we do not analyze cooperation frequencies in the PD in detail because subjects do not have enough opportunity to form their beliefs about punishment levels in the strict one-shot design. Specifically, we did not expect to find differences of cooperation rates between conditions because subjects need some experience to learn the adjustments of punishment behavior reported in this paper. However, this is clearly an important avenue for future research. For the interested reader we report the frequencies here: In the 2PP, 81.8 percent (18 out of 22) cooperated in *M0*, 72.7 percent (48 out of 66) in *M5*, and 76.1 percent (35 out of 46) in *M10*. A Kruskal–Wallis test does not reject the hypothesis that the cooperation rates are equal in the three conditions ($p = 0.6879$). In the 3PP, 72.7 percent (16 out of 22) cooperated in *M0*, 65.2 percent (43 out of 66) in *M5*, and 69.6 percent (32 out of 46) in *M10*. Again, a Kruskal–Wallis test does not reject the hypothesis that the cooperation rates are equal in the three conditions ($p = 0.7716$).

Table 3

Aggregate performance of second and third party punishment at different monitoring fee levels.

Condition	Incentive 2PP	Incentive 3PP Target player's coplayer is		2P overhead Coplayer in 3PP is	
		Cooperator	Defector	Cooperator	Defector
M0	7.0	4.6	1.1	2.4	5.9
M5	2.2 (.0000)	1.3 (.0000)	0.6 (.1354)	0.9 (.2343)	1.6 (.0001)
M10	1.5 (.0000)	1.3 (.0000)	1.1 (.4916)	0.2 (.3548)	0.4 (.0000)

Incentive means punishment of cooperation subtracted from punishment of defection. *SP overhead* means incentive in the 3PP subtracted from incentive in the 2PP. Reported in italics are the p -values of Mann-Whitney rank-sum tests with the H_0 that the respective value is equal to the one in M0. Tests M5 vs M10 yield $p = .7671$ in the fifth column and $p = .2970$ in the sixth column.

conditions pooled together using a Wilcoxon test yields $p = .8185$ (for the two conditions separately $p = .3657$ in M5 and $p = .4795$ in M10). Thus, the decrease of T -types as monitoring gets costly is not significantly different between second and third parties. However, the increase of N -types as monitoring gets costly is greater in second than in third parties: the frequency of N -types is apparently not different between second and third parties in the baseline, but higher in second parties than in third parties when monitoring is costly.²¹ Furthermore, the increase of B -types as monitoring gets costly is greater in third than in second parties: in the baseline, the frequency of B -types is not significantly different from zero in both second and third parties, but strictly positive and higher in third parties than in second parties when monitoring is costly.²² Thus, as a response to monitoring costs, second parties have a slightly stronger tendency to switch to non-punishment while third parties have a slightly stronger tendency to switch to indiscriminate punishment.

Those aggregate differences between the 2PP and the 3PP are driven by individuals that behave differently when they find themselves in the role of a second or third party, respectively. This is shown by the interior cells of Table 2: across all monitoring fee conditions, 27.6 percent (37 out of 134) of the subjects change their behavioral type between the roles. Those subjects are represented by the off-main-diagonal cells in Table 2, where each cell illustrates a different transition path. The key insight from this data is that (i) there are disproportionately more subjects (19) that punish indiscriminately (type B) as a third but not as a second party than the other way around (6 subjects), and that (ii) there are disproportionately more subjects (18) that do not punish at all (type N) as a second but not as a third party than the other way around (7 subjects).

However, the differences leave only a small footprint in the average punishment levels. Return to Table 3 and subtract the incentive to cooperate in the 3PP from the one in the 2PP. We call the resulting value «2P overhead» and show it in the two right-hand side columns.²³ If this overhead is zero, second and third party punishment are equally strong (in the sense of generating the same incentive to cooperate); the larger the overhead, the stronger is second party punishment relative to third party punishment (where a negative value indicates that third party punishment is stronger). The statistics permit two observations: First, all overheads are positive, i.e. second party punishment always generates stronger incentives to cooperate. Second, the overheads are strictly decreasing in the monitoring fee level, i.e. the incentives to cooperate in the 2PP and 3PP, respectively, converge towards equality as the monitoring fee increases. However, only some of the effects are statistically significant.

4. Conclusion

In this paper we investigated how costly second and third party punishment in the context of a social dilemma is affected by the presence of monitoring costs by means of a novel experimental design based on the strategy method. We find that both second and third party punishment is less discriminate and hence generates weaker incentives for cooperation when monitoring is costly compared to a setting in which monitoring is free, because more subjects do not punish at all and some punish unconditionally. There are subtle differences between second and third parties: The presence of monitoring costs leads subjects to withhold sanctioning more often as second parties than as third parties, and to punish indiscriminately more often as third parties than as second parties.

Our results contribute to the understanding of the efficacy of informal sanctioning in resolving social dilemmas. In many important settings persons in a position to mete out punishments are not perfectly informed whether the target player cooperated or defected (see the examples in the introduction). Whether and to what extent such persons remedy the lack

²¹ Comparing the frequency of N -types between the 2PP and the 3PP in both costly monitoring conditions pooled together using a Wilcoxon signed-rank test yields $p = .0164$ (for the two conditions separately $p = .3657$ in M5 and $p = .0114$ in M10). As an alternative demonstration, a dummy indicating treatment (monitoring is costly) is significantly positively correlated with a dummy indicating type N (Kendall's $\tau_b = 0.2995, p = .0006$), whereas this correlation is somewhat stronger for second parties ($\tau_b = 0.3245, p = .0002$) than for third parties ($\tau_b = 0.2421, p = .0053$).

²² Comparing the frequency of B -types between the 2PP and the 3PP in both costly monitoring conditions pooled together using a Wilcoxon test yields $p = .0143$ (for the two conditions separately $p = .0833$ in M5 and M10, respectively). A dummy indicating treatment (monitoring is costly) is significantly positively correlated with a dummy indicating type B (Kendall's $\tau_b = 0.2066, p = .0174$), whereas this correlation is slightly weaker for second parties ($\tau_b = 0.1514, p = .0818$) than for third parties ($\tau_b = 0.1724, p = .0473$).

²³ Note that due to our experimental design harnessing the strategy method and a within-subject comparison of second and third party behavior, this «difference in differences» is identified at the individual level.

of information before deciding to punish, and how they punish under uncertainty has important ramifications for the incentive structure emerging from informal punishment. Our results suggest that lower monitoring costs lead to more efficient outcomes. In practice, any technological and institutional change that increases the transparency of group-relevant behavior (in our experiment: behavior in stage one) effectively lowers monitoring costs. Investigating the long-term effects of such changes on cooperation and efficiency in the lab and the field is a valuable avenue for future research.

We also believe that our results further the understanding whether there is a common motivational structure underpinning both types of punishment or not. So far, the experimental evidence on this question has been less than conclusive. In line with recent behavioral (Leibbrandt & López-Pérez, 2012) and neuroscientific evidence (Buckholtz et al., 2008) our findings show that the responses of second and third party punishment to variations in the cost of monitoring yield similar patterns. This is not to be interpreted as evidence that both types of punishment share an identical motivational structure. But a similar motivational structure would be consistent with this finding.

Finally, we hope that our method to mimic information structures by constraints of the strategy space will be useful to researchers studying other settings with imperfect information within a SM-design.

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Appendices A and B. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.joep.2016.03.002>.

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